Do measures of wealth impact the annual household recycling waste rate in all London boroughs?

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Abstract

Recycling is a vital part of a modern economy and is one of the main environmental issues facing us today since it is necessary for preserving finite natural resources and reducing pollution. Hence, it is crucial to understand what drives recycling to enable us to increase it. Thus, in this study, we examined three potential predictors of recycling rates with a focus on how different measures of wealth could affect recycling rates.

The three measures of wealth that we employed were the mean income, median income and homeownership rate in all London Boroughs between 2006 and 2020, which were independently compared to recycling rates on an individual borough level.

The research found that both mean income and median income did not predict household recycling rates, while outright home ownership explained almost 40 percent of the data's variation ($R^2 = 0.393$). Outright home ownership was initially used as a measure of wealth, assuming that the wealthier a borough, the more people would own their houses outright. However, this was a simplification, and this variable may represent other factors, such as how long a person has lived in an area or the type of housing that is present, with more ownership expected in suburban areas compared to high-density urban areas. Therefore, this paper shows how factors other than wealth could be more accurate predictors of recycling rates, as an area for further study.

Introduction

Recycling is a key issue in London currently, with the Mayor of London setting a target of 50% of all household waste being recycled by 2030. However, currently this figure sits at around 33% for all of London and varies drastically between boroughs. This variation is an important area of study, as understanding why it varies so widely could be key to meeting the 50% target.
There are a multitude of possible predictors that may affect recycling rates since '[they] are the product of a complex interaction of a number of factors' (London Assembly, 2011). However, 'socio-economic factors and housing density are generally perceived to be the main reasons for differences in recycling rates' (London Assembly, 2011). Still, as argued in the London Assembly (2011), these factors are not necessarily the sole predictors. This is why our research aims to examine how different measures of wealth affect the recycling rates of London boroughs, to see if this assumption holds.

The three measures of wealth we will use are mean income, median income and home ownership rates on a borough level with data from 2006-2020. Which is then compared to recycling rates from the same time period using a simple linear regression to determine their R^2 values. We use both mean and median income as measures to minimise the effects of unequal wealth distribution which would skew the mean income, on our findings. Home ownership was used as an alternative measure of wealth and as a potential insight into other factors that may influence recycling rates.
In this paper, we aim to determine whether measures of wealth and property ownership are accurate predictors of recycling rates within London. The paper will first outline our methodology and data sources, followed by a linear regression analysis of our three metrics and concludes by situating the results in a real-world context;

We have the following hypotheses:
For all null hypotheses, we assume that there is a relationship between the independent and dependent variables. While for all alternative hypotheses, it is assumed that there is no relationship between the independent and dependent variables. For all tests an alpha-value of 0.005 will be used to determine statistical significance.
Hence:

- Research question 1:
    - **H_0:** There is **no relationship** between *Mean Income* and *Annual Household Waste Recycling Rate* in all London boroughs.
    - **H_1:** There is **a relationship** between *Mean Income* and *Annual Household Waste Recycling Rate* in all London boroughs.

- Research question 2:
    - **H_0:** There is **no relationship** between *Median Income* and *Annual Household Waste Recycling Rate* in all London boroughs.

- **H_1:** There is **a relationship** between *Median Income* and *Annual Household Waste Recycling Rate* in all London boroughs.

- Research question 3:
  - **H_0:** There is **no relationship** between *House Ownership* and *Annual Household Waste Recycling Rate* in all London boroughs.
  - **H_1:** There is **a relationship** between *House Ownership* and *Annual Household Waste Recycling Rate* in all London boroughs.

Data

This study used secondary quantitative data from the London Datastore - Greater London Authority (London Datastore). We used four data sets, which all included data for all 32 London boroughs between the years 2006 and 2020:

- Average Income of Taxpayers, Survey of Personal Incomes, HMRC
- Housing Tenure, Annual Property Survey (a sample survey)
- Household Waste Recycling Rates, Department of Environment, Food and Rural Affairs (DEFRA)
- Average House Prices, MSOA & LSOA

These data sets enabled us to obtain the following information needed, by borough across 2006-2020:

- Mean and Median Income
- Outright Home Ownership
- Annual Household Recycling Rates (HHRR)
- Mean House Prices of houses in all London Boroughs as a control variable to ensure that Housing Tenure, more specifically, outright home ownership represents wealth rather than areas where it's cheapest to buy a house (less affluent neighbourhoods).

We chose these data sets as they best represent the predictors of wealth we wanted to test alongside our dependent variable, the annual household recycling rates. As we were looking at the recycling rates across London Boroughs, we made sure to choose sets that focused on these areas.

Sampling Methods

All the data collected is secondary in nature. Therefore, it is important to understand how the data was sampled to spot potential biases or faults of the sampling methods employed:

Housing Tenure by Borough:

Sampling survey of 80,000 households (175,000 respondents) split into 5 cohorts. The data has been described as 'indicative only' due to the nature of the sampling survey. The method of choosing the households is unknown.

Average House Prices:

Only properties lodged with HM Land Registry, excluding properties less than £1,000 and greater than £20 million, were included in calculating the average, aiming to mitigate the influence of outliers on the average. Moreover, the geometric mean was used.
The data was cleaned and formatted in a way that allowed all three variables described above onto one table between the years 2006 - 2020 so that our model could be run.

Household Recycling Rate:
Data is supplied from the annual DEFRA Municipal Waste Management Survey, which is collected from local authorities. Local authorities have access to “wastedataflow.org”, where they are given guidance on accurately recording recycling. The recycling rate is calculated as (X/Y) * 100, where X is the tonnage of household waste sent to recycling and Y is the total tonnage of household waste.

Average Income of Taxpayers:
The original data is kept by HMRC and covers all individuals who are liable to pay tax or income that is assessable for tax, therefore excluding non-tax-paying individuals.

As the data is all secondary, there are limitations to it. The entire methodology of the creators of the data sets are largely unknown, but due to the large sample sizes provided, are still useful in our endeavour of understanding what factors of wealth, if any, influence recycling rates in London. Fortunately, as London Datastore is a government-run website, we can assume the data supplied is reliable.

Model

We chose simple linear regression models to test whether factors of wealth, the independent variables, demonstrate a statistically significant relationship with the dependent variable, Household Recycling Rates. Simple linear regressions make the following assumptions:

- There exists a linear relationship between the independent and dependent variable
- The residuals are normally distributed
- Independence of observations: data was collected using appropriate sampling methods and no underlying relationship between the observed.
- Homoscedasticity: the size of the error doesn't significantly change across the independent variable.

In the data section of this report e validity of the sampling methods chosen by the creators of the data sets we chose was discussed and it was concluded that we are confident in the independence of the observations.

We ran initial multilinear models that produced high condition numbers showing our independent variables were potentially collinear or would lead to numerical instability, leading to its exclusion in the model.

The model equation is as follows:

$$Y=\beta_0+\beta_1 X+\varepsilon_i$$

Where:
Y_i = Household Recycling Rate
β_0 = y intercept
β_1 = slope / gradient
X_i = predictor variable
ε_i = error value

And is used to determine how much the household recycling rate is predicted to change for every one unit the predictor value changes.

We ran three individual simple linear regression models:
Recycling ~ mean income - to see if there is a relationship between recycling and mean income
Recycling ~ median income - to see if there is a relationship between recycling and median income
Recycling ~ rate of homeownership - to see if there is a relationship between recycling and homeownership.
Outright Homeownership ~ Average House prices. We used average house prices as a supplementary model to determine whether there existed a statistically significant relationship between it and homeownership (whether homeownership is a measure of wealth).

Results

Median Income

Figure 1.1 - Residual Plot for Household Recycling Rate ~ Median Income

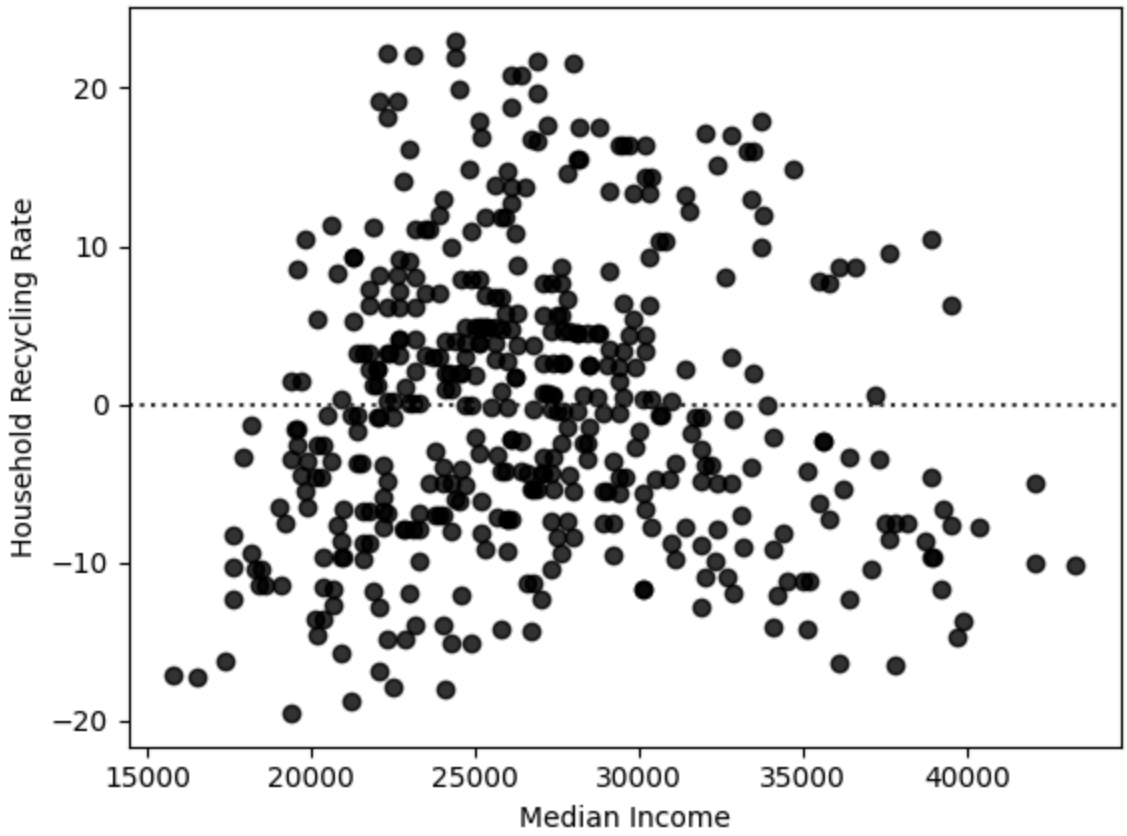

Figure 1.1 shows the residual plot of the relationship between the household recycling rate (HHRR) and the median income of all London Boroughs from 2006 to 2020. It can clearly be observed how this plot is not independent and identically distributed (iid) since the datapoints are arranged in a systematic curvature, rather than clutter around the dotted zero-line with an equal amount above and below, indicating that this model likely is non-linear (McKean, J. W., & Sheather, S. J., 2003). Hence, the R^2-value is unreliable for it fails to capture the true goodness-of-fit due to model misspecification (McKean, J. W., & Sheather, S. J., 2003).

Figure 1.2 - Household Recycling Rate ~ Median Income

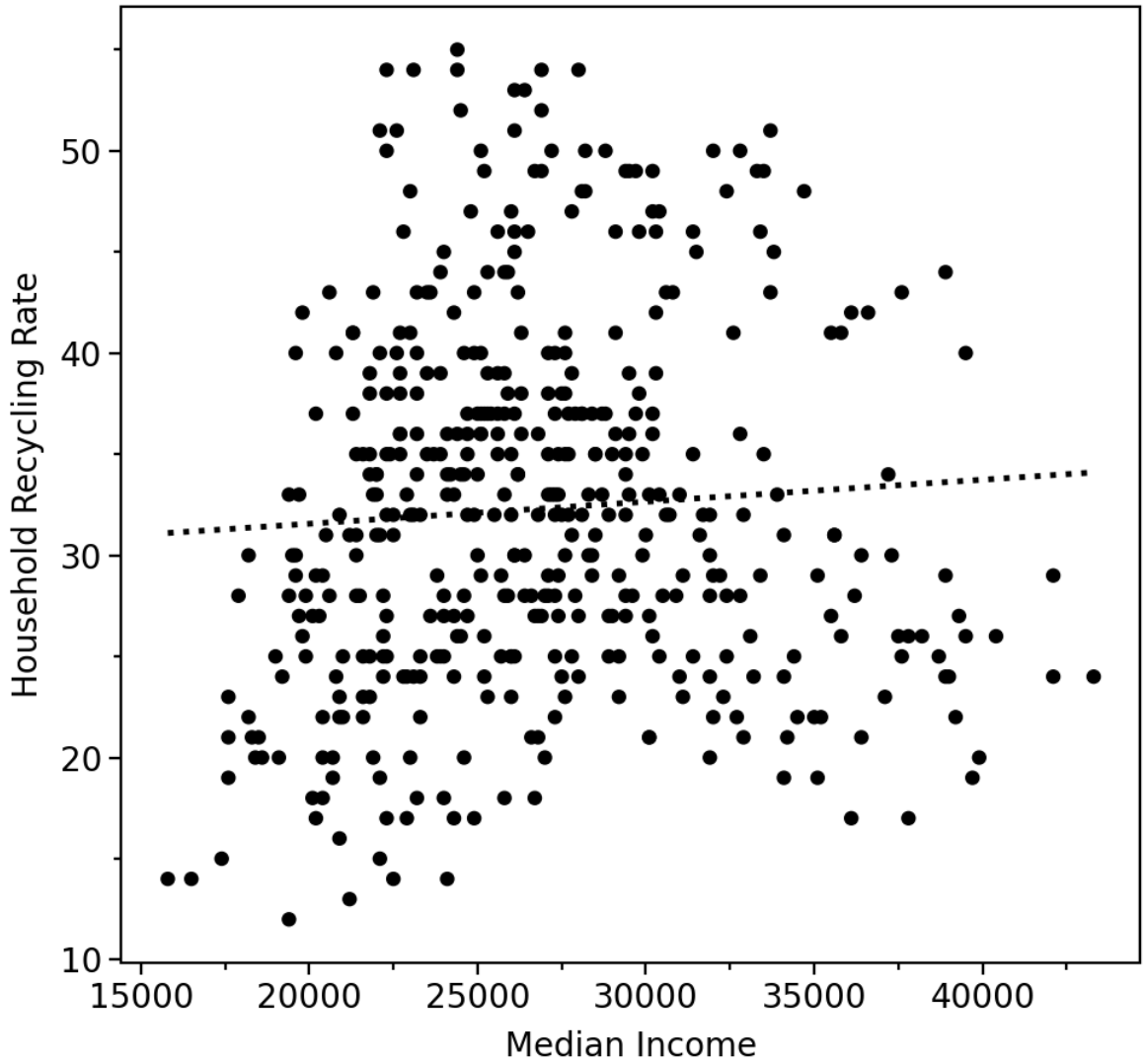

Figure 1.2 illustrates an insignificant weak positive correlation between HHRR and the median income of London Boroughs since the model's line of best fit (dotted line) has a slope only slightly above zero and an R^2-value of 0.004, meaning that only 0.4 percent of the data's variation are explained by the median income. This, accompanied by Figure 1.1, suggests a failure to reject H_0 and thus concludes that there is no relationship between those two variables. However, the systematic curvature observed in Figure 1.1 indicates the absence of a linear relationship, not all relationships. Therefore, there might be a relationship where recycling changes in a non-linear fashion, which our model is mathematically incapable of capturing (McKean, J. W., & Sheather, S. J., 2003). Consequently, using a linear model is inconclusive since it lacks the statistical power to provide a valid conclusion regarding H_0 (Atkinson et al., 1983).

Mean Income

Figure 2.1 - Residual Plot for Household Recycling Rate ~ Mean Income

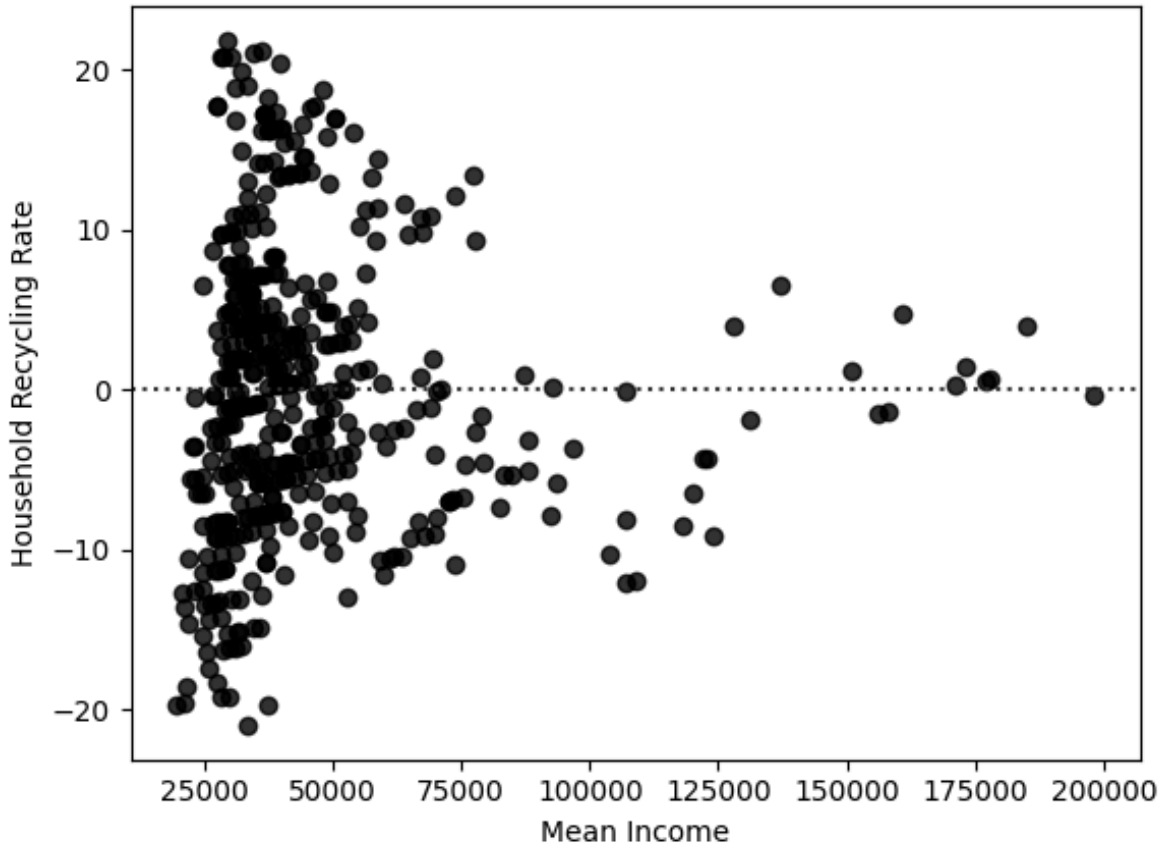

Figure 2.1 shows a non-constant variance, demonstrating the heteroscedasticity of the relationship between the household recycling rate (HHRR) and the median income of all London Boroughs from 2006 to 2020. This violates the model's assumption of homoscedasticity (McKean, J. W., & Sheather, S. J., 2003). Additionally, the model displays a differential predictive accuracy, predicting the HHRR for higher-mean-income boroughs significantly more precisely than for lower-mean-income ones, for which the spread is nearly 40 units (Berk, 2004). Since the R^2-value is a global average, this could further distort its accuracy. However, R^2 is calculated using the sum of squared residuals, and thus, the 40-unit spread observed in the lower-mean-income boroughs drowns this out (Berk, 2004).
Therefore, as for the median income, the R^2-value of this model is unreliable.

Figure 2.2 - Household Recycling Rate ~ Mean Income

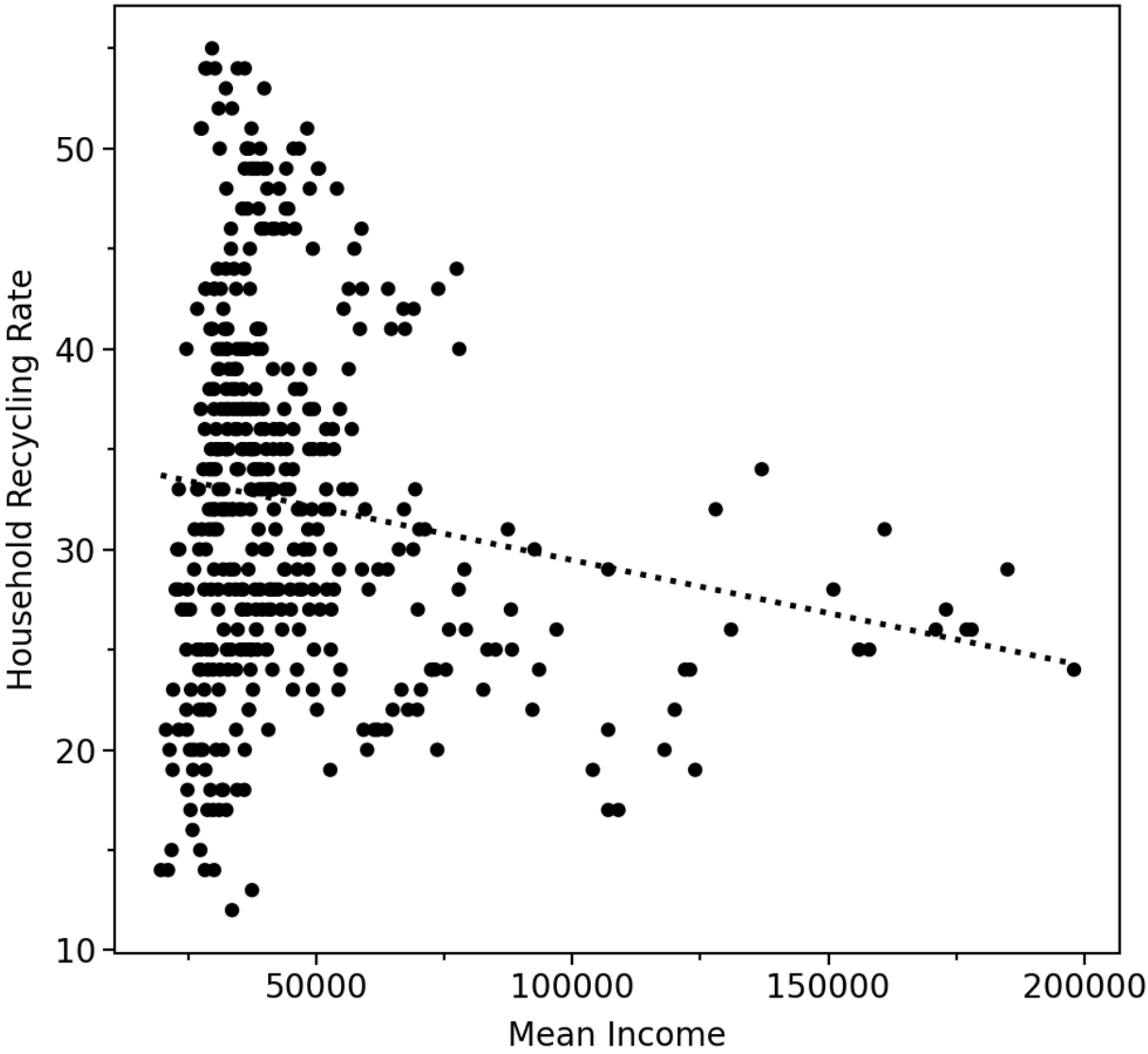

Figure 2.2 highlights the insignificant weak negative correlation between HHRR and the mean income of London Boroughs. This relation is shown by the model's line of best fit (dotted line), which has a slope only slightly below zero and an R^2-value of 0.024, meaning that only 2.4 percent of the data's variation are explained by mean income.

However, regardless of how insignificant the results are, a valid conclusion regarding H_0 cannot be drawn due to the data's heteroscedasticity biasing the standard errors (Long & Ervin, 2000). Therefore, just as for the median income, H_0 can neither be rejected nor fail to reject (Atkinson et al., 1983).

Outright Homeownership

Figure 3.1 - Residual Plot for Household Recycling Rate ~ Outright Homeownership

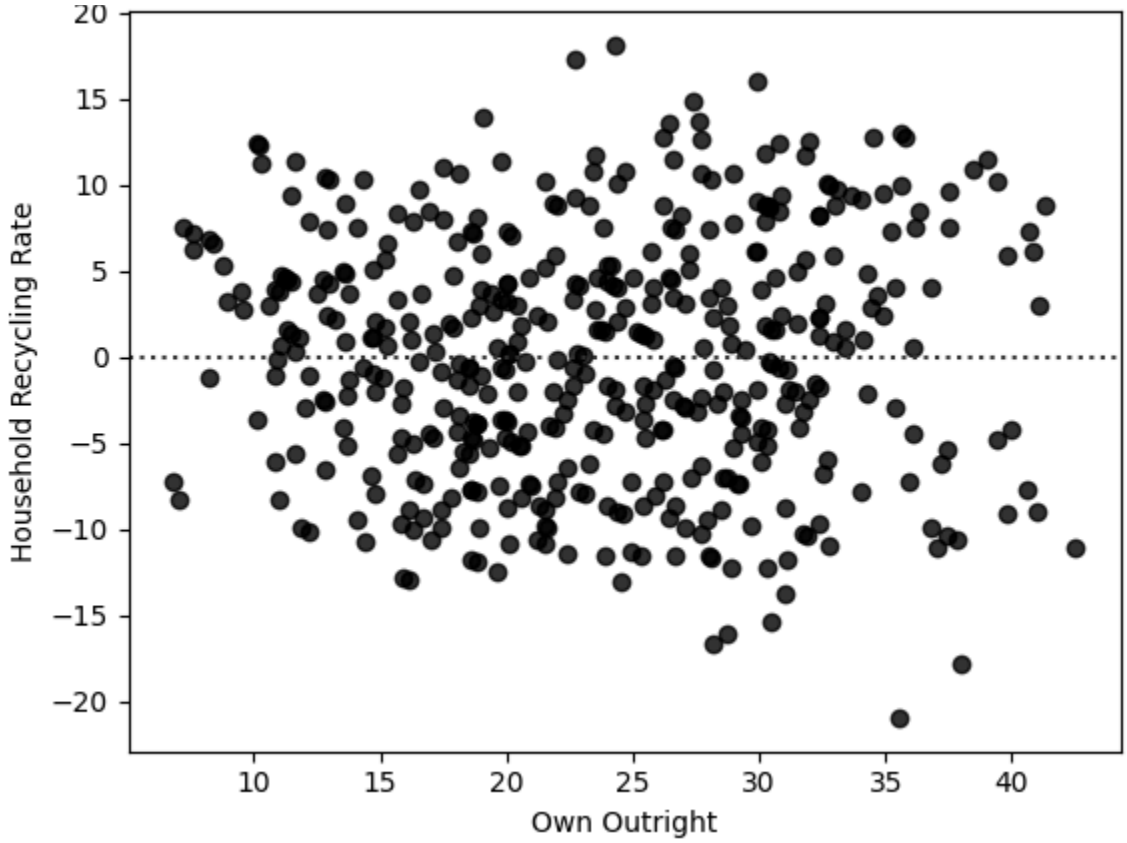

Figure 3.1 displays homoscedasticity, meaning the vertical spread of residuals remains constant across the range of outright homeownership. Hence, unlike Figures 1.1 and 2.1, which represented a “C-shaped” and a “fan-shaped” distribution, this residual plot illustrates a “random noise” distribution (McKean, J. W., & Sheather, S. J., 2003). This indicates a random scatter and constant variance, and thus, the R^2-value of this model is reliable.

Figure 3.2 - Household Recycling Rate ~ Outright Homeownership

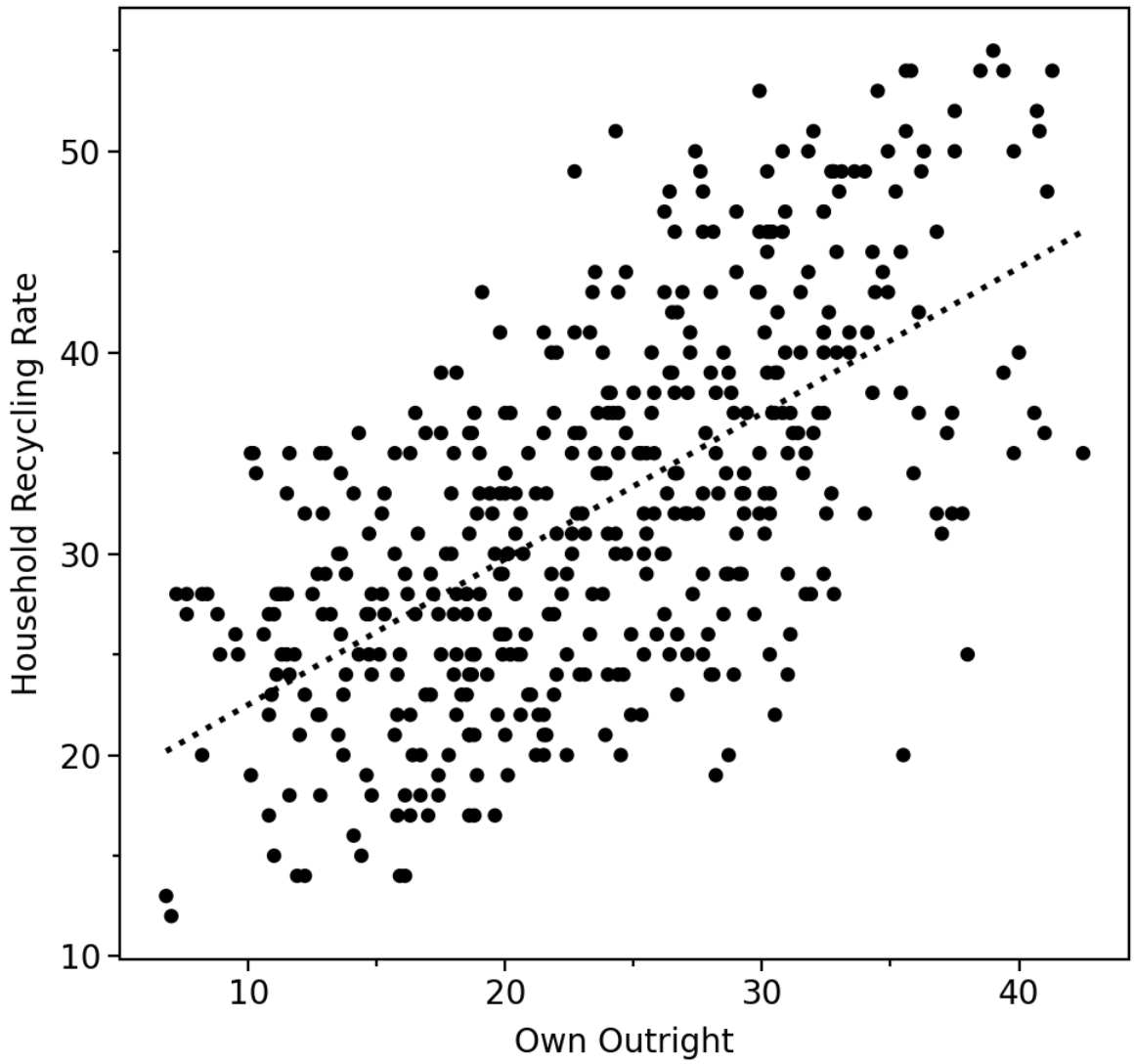

Figure 3.2 illustrates a significant positive relationship between HHRR and outright homeownership, with an R^2-value of 0.393. This indicates that the model accounts for 39.3% of the observed variance in recycling rates. Given that the diagnostic residuals in Figure 3.1 satisfy the i.i.d. assumption, this R^2-value represents a reliable and statistically sound measure of the predictive strength of homeownership status. Hence, a valid conclusion regarding H_0 can be drawn due to the data's homoscedasticity (McKean, J. W., & Sheather, S. J., 2003);
Since the p-value is 2.81e-50, which is smaller than the alpha value of 0.05, there is sufficient evidence to reject $H_{0,}$ and thus, there is a relationship between HHRR and outright homeownership.

Discussions

Both the mean and median income showed no statistically significant results, and both their residual plots indicated that even if they did, their R^2-value would be insignificant. Therefore, this suggests, as touched upon in the introduction, that socio-economic factors might not be the sole or even main predictor for recycling rates. However, our third measure of wealth, outright homeownership, yielded a statistically significant result and an R^2-value of 0.393.
Regardless, this seems odd since our predictors supposedly measure wealth within a borough, and therefore, should yield similar results. Hence, an area with high homeownership should have a high mean or median income since it's wealthy, and all predictors should reflect that. However, this is not the case, indicating that some of these measures might not represent the wealth of a borough, but rather other factors that, as shown through Figure 2.1's extreme variance, are likely to impact HHRR. As argued in Chiripanhura (2011), using the median in addition to the mean is a good way of representing wealth since it accounts for the top few percent's impact on the mean. Additionally, a simple linear regression was performed to measure the strength of the relationship between outright homeownership and house prices, which yielded an insignificant R^2-value of 0.002. Therefore, while homeownership might also measure wealth to an extent, it is more likely to represent different factors, such as the type of housing, how long a person has lived in an area and how much they care about that area, explaining why this was the only predictor with statistical significance and a high R^2-value. Thus, our research indicates that measures of wealth were not the main drivers of recycling in all London boroughs from 2006 to 2020.
However, this is a simplification since only tests for linear relationships were conducted, and as shown in Figure 1.1, the relationship between median income and HHRR is likely non-linear. Hence, to discover the best predictors for HHRR, it would be beneficial to conduct a non-linear test in future research.

Appendix

Design table

Research Question	Hypothesis	Sampling Plan	Analysis	Interpretation
Does **income/wealth** impact the **annual household waste recycling rate** in all London boroughs? RQ1 = Does **mean income** relate to annual household recycling in all London Boroughs from 2006 to 2020? RQ2 = Does **median income** relate to annual household recycling in all London Boroughs from 2006 to 2020? RQ3 = Does **home ownership** (Owing Outright percentage) relate to annual household recycling in all London Boroughs from 2006 to 2020?	**H_0:** There is **no relationship** between mean income (RQ1)/median income (RQ2)/outright home ownership (RQ3) and the annual household waste recycling rate in all London Boroughs from 2006 to 2020. **H_1:** There is **a relationship** between mean income (RQ1)/median income (RQ2)/outright home ownership (RQ3) and the annual household waste recycling rate in all London Boroughs from 2006 to 2020.	Data was obtained from the Data Store London from the UK government website, which we will **Mean income:** A decent measure of the wealth of a borough **Median Income:** The mean income can be distorted by the top earners, thus the median could give a more accurate representation of a borough's waste **Home Ownership:** (Owing Outright percentage)**:** Wanted to use different measures of wealth, and used outright home ownership → Besides measuring wealth in another way, this could also reflect	Use of linear regression, looking at each predictor individually to determine which, if any, predicts the rate of recycling in a borough.	**RQ1:** A positive correlation between the IV and the DV would indicate that higher-income boroughs have better recycling habits. A negative correlation between the IV and the DV would indicate that lower-income boroughs have better recycling habits. **RQ2:** A positive correlation between the IV and the DV would indicate that within higher-income boroughs, the majority of households have a higher recycling rate. A negative correlation between the IV and the DV would indicate that within higher-income boroughs, the majority of

		how connected a person is to their neighborhood, since if you own a house, you're usually planning on staying in that neighborhood		households have a lower recycling rate. **RQ3:** A positive correlation between the IV and DV would indicate that boroughs with a higher house ownership have a higher recycling rate. A negative correlation between the IV and DV would indicate that boroughs with a higher house ownership have a lower recycling rate.

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